

regardless of frequency.

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$$E = \frac{p^2}{2m} = \frac{\left(\frac{h}{\lambda}\right)^2}{2m} = 12.31 \text{ eV}$$

The electron can lose part of its kinetic energy and will result in excitation to -1.51 eV .

Note: Option D is not the answer because the photon will not be absorbed as its energy is not exactly equal to the difference between the energy levels.

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$$\Delta x \Delta p \geq h$$

Since $\Delta x = 3.0 \times 10^{-10} \text{ m}$